实验原理公式汇总

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1 词嵌入

$$\mathbf{e}_i = E \cdot \text{onehot}(t_i), \quad \mathbf{e}_i \in \mathbb{R}^{d_{model}} \tag{1}$$

2 位置编码

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right) \tag{2}$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right) \tag{3}$$

$$\mathbf{x}_i = \mathbf{e}_i + PE_{pos=i} \tag{4}$$

3 自注意力机制

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \tag{5}$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{6}$$

4 多头注意力

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W_O \tag{7}$$

$$\text{head}_j = \text{Attention}(QW_Q^j, KW_K^j, VW_V^j) \tag{8}$$

5 前馈网络与残差连接

$$\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \tag{9}$$

$$x' = \mathrm{LayerNorm}(x + \mathrm{SubLayer}(x)) \tag{10}$$

6 分类头与损失函数

$$\mathbf{h} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i^{(L)} \tag{11}$$

$$\mathbf{z} = \mathbf{h}W_{cls} + b_{cls}, \quad \mathbf{z} \in \mathbb{R}^2 \tag{12}$$

$$L = -\sum_{c=1}^{2} y_c \log\left(\frac{e^{z_c}}{\sum_{k=1}^{2} e^{z_k}}\right) \tag{13}$$

7 优化与正则化

$$\theta \leftarrow \theta - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} - \eta\lambda\theta \tag{14}$$